

# Nate Cirino

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Available Jan 2027

## EDUCATION

### Northeastern University

Aug 2027 | MGPA: 3.6/4.0

*B.S. in Computer Science, Systems Concentration*

*Boston, MA*

**Coursework:** Computer Systems, Computer Networks, Computer Architecture, Algorithms & Data Structures, Object-Oriented Design, Foundations of Software Engineering, Systems Security, Cybersecurity, Data Science

**Honors:** Dean's List (all semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor

## EXPERIENCE

### Wayfair

Jul 2026 – Dec 2026

*Software Engineering Co-op*

*Boston, MA*

- Building a REST API that unifies three competing e-label suppliers (Vusion, Pricer, Solum) behind one interface, enabling dynamic price updates across 50,000+ shelf labels and streamlining new outlet launches
- Operate 500+ front-of-house POS and e-label devices at 99.9% uptime with sub-3-minute average resolution across all four Wayfair retail brands (AllModern, Birch Lane, Joss & Main, Perigold)

### Wolters Kluwer

Jul 2025 – Jun 2026

*Software Engineer*

*Boston, MA*

- Overhauled JSON serialization across 20+ microservices with json-io, reducing average payload size 35% and cutting inter-service communication latency 28% during high-traffic clinical query periods
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing access for 10K+ clinicians with zero degradation during peak EHR synchronization windows
- Sustained 92% uptime for the Expert AI inference pipeline serving 250K+ daily active clinicians

### SculptAI

Jan 2025 – Aug 2026

*Software Architect*

*New York, NY*

- Implemented a FAISS in-memory vector store with periodic re-indexing and AWS S3 Vectors for persistence, holding retrieval p95 under 150ms at peak while cutting vector storage cost roughly 70% vs. OpenSearch Serverless
- Built Sentinel, an AI SRE agent in production, cutting MTTR on P0/P1 incidents from 45 to 12 minutes and reducing MTTD 60% by correlating Prometheus, Tempo, and Loki telemetry via tool-use
- Re-orged 20+ engineers into Retrieval, Generation, and Observability squads; now technical advisor

### Chevron New Energies

May 2024 – Aug 2024

*Software Engineer Intern*

*Houston, TX*

- Orchestrated 200+ Azure Data Factory pipelines across 10+ data sources at a 99.95% success rate, holding retry-induced latency overhead under 1% via exponential-backoff retry policies
- Reduced change failure rate from 15% to 2% with multi-stage Azure DevOps pipelines running automated unit, integration, and performance tests in staging to catch regressions before production

## PROJECTS

### Kepler | Rust, C++, Raft, LSM-tree, MVCC, Tokio

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and an LSM-tree engine
- Reduced API server latency 25% and held 99% uptime across 100 pods under write-heavy load; eliminated managed-etcd dependence via an MVCC layer, cutting infra cost \$2,500/month and raising TPS headroom 30%

### NOAA AUV Sonar Detection | Python, PyTorch, AWS Neuron SDK, SageMaker

Apr. 2025 – Jul. 2025

- Optimized CNN inference to sub-50ms p99 on a single AWS Inferentia2 core for real-time onboard AUVs
- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures, reaching 94.7% precision and 91.2% recall on a 10,000+ event test set while cutting false positives 75% over prior baselines

## SKILLS & INTERESTS

**Languages & Tools:** Rust, Go, Python, C++, C, Java, TypeScript, JavaScript, Kotlin, Swift, Bash, SQL, PostgreSQL, MySQL, MongoDB, gRPC, GraphQL, Linux, Docker, Kubernetes, Terraform, Buildkite, AWS, Azure, GCP

**Frameworks:** Spring Boot, FastAPI, Flask, Express, Node.js, React, Next.js, PyTorch, TensorFlow

**AI / Agentic Tooling:** Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

**Interests:** Avid Cleveland Browns fan, Weightlifting, Rock Climbing, BJJ, Running